AI-Anamnesis of Patients

Evaluating End-to-End STT + LLM Pipelines for Medical Documentation

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Master Seminar in Cooperation with PwC Cloud & AI · Hochschule Hannover · 2026

The Problem

2 hours per day — that's how much time physicians spend on documentation for every hour of patient care (Sinsky et al., Annals of Internal Medicine, 2016). 49% of physicians report burnout, with administrative load as the #1 driver (Medscape, 2024). Our project evaluates whether AI can automate this documentation without sacrificing accuracy.

The Pipeline — Four Stages

Our end-to-end workflow transforms a doctor-patient audio recording into a structured SOAP note in four stages:

① **Audio Capture —** Recording of the consultation.

② **STT + Diarization —** Speech-to-Text converts audio to text and separates speakers (SPEAKER_A, SPEAKER_B).

③ **Format LLM —** An LLM replaces anonymous speaker labels with 'Arzt' and the patient's name. No other changes.

④ **SOAP LLM —** A second LLM generates the structured medical note: Subjective, Objective, Assessment, Plan.

Tested Combinations

Five model combinations tested across cloud and local deployment:

#	STT	LLM	Type
1	AssemblyAI	GPT-4o	☁️ Cloud
2	Speechmatics	GPT-4o	☁️ Cloud
3	Whisper turbo	SauerkrautLM 8B	▣ Local
4	Whisper turbo	Llama 3.2 3B	▣ Local
5	Whisper turbo	Gemma 4 4B	▣ Local

Hardware constraint: local models ran on a consumer laptop (AMD CPU, 16 GB RAM). Larger models (Whisper large-v3, SauerkrautLM 70B) exceeded available resources.

Test Scenarios — 5 × 11 = 55 Runs

Eleven scenarios in four difficulty tiers, with manually annotated ground truths including hesitations, self-corrections, and verbal slips:

Standard: Original (clean dictation), Laptop-Middle, Laptop-Doctor

Challenging: White Noise, Self-Corrections, Interruptions

Complex: Topic Jumps, Mind Changes, Chaos (all combined)

Real-World: Long Anamnesis (19 min), Physiotherapy (PwC recording)

Evaluation — Three Independent Axes

WER (Word Error Rate) — manual ground truth vs. RAW STT output. Measures transcription accuracy through insertions, deletions, and substitutions.

LLM Fidelity — RAW STT vs. formatted output (after removing expected speaker label changes). Any remaining difference = LLM error.

SOAP Quality — Claude-as-Judge scoring on a 0–8 scale (S + O + A + P, each 0–2). Strict rule: any hallucinated diagnosis or treatment = 0 points for that section.

SOAP is the standard format for medical documentation: Subjective, Objective, Assessment, Plan. Each section covers a different part of the consultation, from the patient's own description to the treatment plan.

Why Claude-as-Judge? 55 runs × 4 sections = 220 evaluations — too many for manual scoring in the timeframe, and script-based evaluation cannot assess medical semantic correctness. Claude provides consistent scoring with a documented evaluation prompt. Claude was not used as a model in our pipeline, which eliminates any self-evaluation bias. Had we used GPT-4o as the judge, it would likely have favored its own outputs.

Key Results

1. Transcription Quality (WER)

Model	Avg WER	Type
Assembly	**8.9 %**	☁️ Cloud
Speechmatics	11.4 %	☁️ Cloud
Whisper (local avg)	13.5 %	▣ Local

Critical finding: In the White Noise scenario, all models collapse — Assembly reaches 33.2%, Speechmatics 30.6% and Whisper-local 41%. Microphone quality sets the floor.

2. LLM Fidelity — Formatting Errors

Model	Avg Error Rate	Type
GPT-4o (cloud avg)	**0.0 %**	☁️ Cloud
Gemma	13.3 %	▣ Local
Llama	15.1 %	▣ Local
Sauerkraut	19.7 %	▣ Local

The Anamnese Catastrophe: SauerkrautLM reduced a 19-minute conversation (2,269 words) to just 152 words — a 93% silent content loss. All local models struggle severely with long conversations due to limited context windows.

3. SOAP Quality — Clinical Usability

Model	Ø Score	+ (6+)	o (3-6)	- (0-3)	Type
GPT-4o (avg)	**4.4 / 8**	0	10	1	☁️ Cloud
Gemma	3.4 / 8	0	5	6	▣ Local
Sauerkraut	3.2 / 8	0	4	7	▣ Local
Llama	2.1 / 8	0	2	9	▣ Local

Zero of 55 runs reached 'acceptable' quality (>6/8). The Plan section averaged 0.0–0.3 / 2 across all models — every system hallucinated treatments.

Verdict

Winners: **AssemblyAI & GPT-4o** — lowest WER (8.9%), zero hallucination in formatting, best SOAP score (4.4/8). But: still not clinically usable. No run reached the acceptable threshold, and the Plan section remains the primary hallucination source.

Privacy — Sovereignty is a Spectrum

Both AssemblyAI and GPT-4o can be hosted in EU-only regions, making the cloud pipeline GDPR-compliant without transatlantic data transfer — a significant step forward. However, EU-hosted is not self-hosted: patient data still flows to a third-party provider. Many hospitals and patients will not accept this, even within EU jurisdiction. The only path to full data sovereignty remains a fully on-premise stack. The ideal deployment — high quality AND no third party — is still the goal that the next steps target.

Deployment	Data Sovereignty	SOAP Score
Fully on-premise *Whisper turbo + local LLMs*	No third party — full control	⌀ 2.9 / 8
EU Cloud *Assembly / Speechmatics + GPT-4o*	EU-hosted, GDPR-compliant — third-party server	⌀ 4.4 / 8
On-prem at scale — target *Whisper large-v3 + SauerkrautLM 70B*	No third party — full control	*>6*

Next Steps

1. Scale up the local stack — Deploy Whisper large-v3 + SauerkrautLM 70B on dedicated GPU infrastructure. Goal: close the quality gap while maintaining data sovereignty.

2. Prompt engineering for better SOAP quality — Constrained generation, 'cite-or-skip' rules. Additionally: chunking for long conversations.

3. Real-time streaming UX — Stream STT during the consultation, generate SOAP on-the-fly. Target: draft note ready before the patient leaves the room.

Three Things to Remember

1. The pipeline works — the Plan section doesn't.
2. Hardware and model size are the bottleneck.
3. Assistive today. Autonomous tomorrow?

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